Bilinear TVP *

Abstract

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1 Proposal

Let $\mathbf{y}_t$ denote an N-dimensional vector of endogenous variables. Without loss of generality, we consider a homoskedastic TVP-VAR(1) model without an intercept:

$$\mathbf{y}_t = \mathbf{\Phi}_t \mathbf{y}_{t-1} + \boldsymbol{\epsilon}_t, \quad \boldsymbol{\epsilon}_t \sim \mathcal{N}(\mathbf{0}, \mathbf{\Sigma}) \tag{1}$$

where each $\mathbf{\Phi}_t$ is a $N \times N$ matrix of lagged TV coefficients. For convenience, we adopt the non-centered parameterization, decomposing TV coefficients into a static ($\bar{\mathbf{\Phi}}$), and a pure TV ($\tilde{\mathbf{\Phi}}_t$) component:

$$\mathbf{y}_t = \bar{\mathbf{\Phi}} \mathbf{y}_{t-1} + \tilde{\mathbf{\Phi}}_t \mathbf{y}_{t-1} + \boldsymbol{\epsilon}_t, \tag{2}$$

so that $\mathbf{\Phi}_t = \bar{\mathbf{\Phi}} + \tilde{\mathbf{\Phi}}_t$, with $\tilde{\mathbf{\Phi}}_0 = \mathbf{0}$.

A fully unrestricted specification involves N^2 TV parameters, which are estimated by vectorizing $\mathbf{A}_t$ and working on the resulting state vector. This approach ignores the intrinsic matrix structure of the coefficient matrix, which can be exploited to reduce the dimensionality of the parameter space by leveraging its two-dimensional nature.

The premise is that, for most lagged coefficients, time variation is either absent or not pervasive. Accordingly, we aim to reduce the total number of latent states. Since $\tilde{\mathbf{\Phi}}_t$ is a matrix, its bilinear structure can be exploited to decompose time variation into row-wise and column-wise components. Specifically, each row i captures the effects of all lagged variables on variable i, while each column j reflects the impact of lagged variable j across all equations. These two dimensions may naturally display distinct patterns of time variation, which can be efficiently captured using a reduced number of parameters.

Let $a_{i,t} \in \mathbf{a}_t$ and $c_{i,t} \in \mathbf{c}_t$ denote TV auxiliary parameters, each evolving according to RW dynamics:

$$\begin{aligned} a_{i,t} &= a_{i,t-1} + v^a_{i,t}, \quad v^a_{i,t} \sim \mathcal{N}(0, s^a_i), \\ c_{i,t} &= c_{i,t-1} + v^c_{i,t}, \quad v^c_{i,t} \sim \mathcal{N}(0, s^c_i). \end{aligned} \tag{3}$$

We approximate each element $\tilde{\phi}_{ij,t}$ such that time variation is captured by these low-dimensional components, representing row-wise and column-wise dynamics:

$$\tilde{\phi}_{ij,t} \approx \hat{\phi}_{ij,t} = a_{i,t}\, b_{ij}\, c_{j,t}, \tag{4}$$

where b_{ij} are static parameters. To ensure that the right-hand side satisfies the initial conditions, we impose $a_{i,0} = 0$ and $c_{j,0} = 1$ (or vice versa, or both at 0, ...). In matrix form, this decomposition can be written as

$$\tilde{\mathbf{\Phi}}_t \approx \hat{\mathbf{\Phi}}_t = \mathbf{A}_t \mathbf{B} \mathbf{C}_t, \tag{5}$$

where $\mathbf{A}_t = \text{diag}(\mathbf{a}_t)$, $\mathbf{C}_t = \text{diag}(\mathbf{c}_t)$, and $b_{ij} \in \mathbf{B}$. An equivalent representation is given by

$$\hat{\mathbf{\Phi}}_t = \mathbf{B} \odot (\mathbf{a}_t \mathbf{c}_t'), \tag{6}$$

which highlights that all time variation is driven by the reduced-rank outer product $\mathbf{a}_t \mathbf{c}_t'$, yielding a low-dimensional representation of the dynamic structure.

1.1 Identification

The auxiliary approximation is based on the product of three latent components, which gives rise to both TV and static identification issues.

- **TV**. For the pair of TV auxiliary coefficients, there may exist time-dependent scalars ν_t such that, for example, $a_{i,t} c_{j,t} = \bar{a}_{i,t} \bar{c}_{j,t}$, where $\bar{a}_{i,t} = \nu_t a_{i,t}$ and $\bar{c}_{j,t} = c_{j,t}/\nu_t$. This source of indeterminacy is automatically ruled out by assuming constant innovation variances in the RW dynamics.
- **Static**. Static identification is more subtle as it also involves the static parameter b_{ij}. In particular, any pair within the triplet may be subject to scale indeterminacy, for instance, $a_{i,t} b_{ij} = \bar{a}_{i,t} \bar{b}_{ij}$, with $\bar{a}_{i,t} = \nu a_{i,t}$ and $\bar{b}_{ij} = b_{ij}/\nu$. To eliminate this, we impose fixed RW innovation variances. For simplicity, we set $s_i^a = s_j^c = 1$.

1.2 Mismatch in variance growth rates

For the unrestricted coefficients,

$$\mathbb{E}[\tilde{\phi}_{ij,t}] = 0, \quad \mathbb{V}[\tilde{\phi}_{ij,t}] = t s_{ij}^{\phi}. \tag{7}$$

For the proposed approximation, under the hypothese of diuble RW dynamics:

$$\mathbb{E}[\hat{\phi}_{ij,t}] = b_{ij} \mathbb{E}[a_{i,t} c_{j,t}] = 0. \tag{8}$$

and

$$\mathbb{V}[\hat{\phi}_{ij,t}] = b_{ij}^2 \mathbb{V}[a_{i,t} c_{j,t}] = t^2 b_{ij}^2 s_i^a s_j^c. \tag{9}$$

The auxiliary variance grows quadratically in t. Assume $a_{i,t}$ is AR(1):

$$a_{i,t} = \phi a_{i,t-1}, \quad |\phi| < 1. \tag{10}$$

Then

$$\mathbb{V}[a_{i,t}] = s_i^a \frac{1 - \phi^{2t}}{1 - \phi^2}. \tag{11}$$

It follows that

$$\lim_{t \to \infty} \frac{\mathbb{V}[\tilde{\phi}_{ij,t}]}{\mathbb{V}[\hat{\phi}_{ij,t}]} = \frac{s_{ij}^{\phi}(1 - \phi^2)}{b_{ij}^2 s_i^a s_j^c}. \tag{12}$$

To ensure stability in finite samples, select ϕ and s_i^a so that

$$s_i^a(1-\phi^{2T})(1-\phi^2)^{-1} = T^{1/2}, \quad s_i^c = T^{-1/2}.$$

Then

$$\mathbb{V}[\hat{\phi}_{ij,T}] = \frac{Tb_{ij}^2 s_i^a(1-\phi^{2T})s_j^c}{1-\phi^2} = T. \tag{13}$$

The issue with this choice is that it introduces an asymmetry between the left and right auxiliary parameters. One potential solution is to set the stationary one to be highly persistent, so that in finite samples it effectively behaves like a RW. A value such as $\phi = 0.99$ may be suitable. [1]

1.3 Sampler

We outline the sampler for the general lag P case, which can be written in non-centered parameterization as

$$\mathbf{y}_t = \bar{\mathbf{\Phi}}\mathbf{x}_{t-1} + \tilde{\mathbf{\Phi}}_t\mathbf{x}_{t-1} + \boldsymbol{\epsilon}_t, \tag{14}$$

where $\bar{\mathbf{\Phi}} = [\bar{\mathbf{\Phi}}_1, ..., \bar{\mathbf{\Phi}}_P]$, $\bar{\mathbf{\Phi}}_t = [\bar{\mathbf{\Phi}}_{1,t}, ..., \bar{\mathbf{\Phi}}_{P,t}]$. The proposed decomposition is applied independently to each lag as

$$\hat{\mathbf{\Phi}}_{p,t} = \mathbf{A}_{p,t}\mathbf{B}_p\mathbf{C}_{p,t} \tag{15}$$

where $\mathbf{A}_{p,t} = \text{diag}(\mathbf{a}_{p,t})$, and $\mathbf{C}_{p,t} = \text{diag}(\mathbf{c}_{p,t})$. Moreover, let $\mathbf{b}_p = \text{vec}(\mathbf{B}_p)$, $\mathbf{b} = \text{vec}([\mathbf{b}_1', ..., \mathbf{b}_P'])$, $\mathbf{a}_t = [\mathbf{a}_{1,t}', ..., \mathbf{a}_{P,t}']'$, $\mathbf{c}_t = [\mathbf{c}_{1,t}', ..., \mathbf{c}_{P,t}']'$

1. **$\bar{\mathbf{\Phi}}$ step**. Conditional on the components of the TV decomposition, sampling $\bar{\mathbf{\Phi}}$ is straightforward and amounts to a standard Bayesian static VAR update.

2. **b step.** We can isolate $\mathbf{b}$ as

$$\bar{\mathbf{y}}_t = \bar{\mathbf{Y}}_{t-1}\mathbf{b}_1 + ... + \bar{\mathbf{Y}}_{t-P}\mathbf{b}_P + \boldsymbol{\epsilon}_t = \bar{\mathbf{X}}_{t-1}\mathbf{b} + \boldsymbol{\epsilon}_t \tag{16}$$

   where $\bar{\mathbf{y}}_t = \mathbf{y}_t - \bar{\mathbf{\Phi}}\mathbf{x}_{t-1}$, $\bar{\mathbf{Y}}_{t-p} = (\mathbf{y}_{t-p}'\mathbf{C}_{p,t} \otimes \mathbf{A}_{p,t})$, $\bar{\mathbf{X}}_{t-1} = [\bar{\mathbf{Y}}_{t-1}, ..., \bar{\mathbf{Y}}_{t-P}]$.

3. **$\mathbf{a}_t$ step.** We can isolate $\mathbf{a}_t$ as

$$\bar{\mathbf{y}}_t = \tilde{\mathbf{Y}}_{t-1}\mathbf{a}_{1,t} + ... + \tilde{\mathbf{Y}}_{t-p}\mathbf{a}_{p,t} + \boldsymbol{\epsilon}_t = \tilde{\mathbf{X}}_{t-1}\mathbf{a}_t \tag{17}$$

   where $\tilde{\mathbf{Y}}_{t-p} = \text{diag}(\mathbf{B}_p\mathbf{C}_{p,t}\mathbf{y}_{t-p})$, $\tilde{\mathbf{X}}_{t-1} = [\tilde{\mathbf{Y}}_{t-1}, ..., \tilde{\mathbf{Y}}_{t-P}]$.

[1] Similar issues arise in Del Negro and Otrok (2008); Stock and Watson (2016).

4. $\mathbf{c}_t$ **step.** We can isolate $\mathbf{c}_t$ as

$$\bar{\mathbf{y}}_t = \hat{\mathbf{Y}}_{t-1}\mathbf{c}_{1,t} + ... + \hat{\mathbf{Y}}_{t-P}\mathbf{c}_{P,t} + \boldsymbol{\epsilon}_t = \hat{\mathbf{X}}_{t-1}\mathbf{c}_t + \boldsymbol{\epsilon}_t \tag{18}$$

where $\hat{\mathbf{Y}}_{t-p} = \mathbf{A}_t\mathbf{B}\text{diag}(\mathbf{y}_{t-p})$.

2 Modeling TV covariance matrices

The same idea can be applied in the context of modeling TV covariance matrices. A convenient approach to tackle the nonlinear problem of modeling TV covariances is to "linearize" it via a factor decomposition. Let $\mathbf{f}_t$ be an R-dimensional vector of factors. A general TV factor decomposition of $\text{Cov}(\boldsymbol{\epsilon}_t) = \boldsymbol{\Sigma}_t$ can be written as:

$$\boldsymbol{\epsilon}_t = \boldsymbol{\Lambda}_t\mathbf{f}_t + \mathbf{u}_t, \tag{19}$$

where

$$\begin{bmatrix} \mathbf{f}_t \\ \mathbf{u}_t \end{bmatrix} \sim \mathcal{N}\left(\mathbf{0}, \begin{bmatrix} \boldsymbol{\Omega}_t^f & \mathbf{0} \\ \mathbf{0} & \boldsymbol{\Omega}_t^u \end{bmatrix}\right) \tag{20}$$

with $\boldsymbol{\Omega}_t^f = \text{diag}(\boldsymbol{\omega}_t^f)$, $\boldsymbol{\Omega}_t^u = \text{diag}(\boldsymbol{\omega}_t^u)$, such that

$$\boldsymbol{\Sigma}_t \approx \boldsymbol{\Lambda}_t\boldsymbol{\Omega}_t^f\boldsymbol{\Lambda}_t' + \boldsymbol{\Omega}_t^u \tag{21}$$

Kastner (2019) assumes constant loadings $\boldsymbol{\Lambda}_t = \boldsymbol{\Lambda}$, whereas Korobilis (2025), in a structural framework, allow for TV (sign restricted) loadings but set $\boldsymbol{\omega}_t^f = \boldsymbol{\iota}_R$.

The proposed decomposition can be applied to $\boldsymbol{\Lambda}_t$, providing a parsimonious generalization of Kastner (2019). At the same time, it emerges as a constrained alternative to a full TV specification (where all loadings are TV) which, to the best of my knowledge, does not exist. This approach effectively reduces the number of TV loadings from RN to $R + N$, and the total number of TV states from $(R + 2)N$ to $R + 3N$.

$$\begin{aligned} \boldsymbol{\epsilon}_t &= \boldsymbol{\Lambda}_t\mathbf{f}_t + \mathbf{u}_t^{\epsilon}, \\ \boldsymbol{\Lambda}_t &= \mathbf{B}_t\mathbf{C}\mathbf{D}_t \end{aligned} \tag{22}$$

with $\mathbf{B}_t = \text{diag}(\mathbf{b}_t)$, $\mathbf{D}_t = \text{diag}(\mathbf{d}_t)$, and

$$\begin{bmatrix} \mathbf{b}_t \\ \mathbf{d}_t \\ \mathbf{f}_t \\ \mathbf{u}_t \end{bmatrix} \sim \mathcal{N}\left(\begin{bmatrix} \mathbf{b}_{t-1} \\ \mathbf{d}_{t-1} \\ \mathbf{0} \\ \mathbf{0} \end{bmatrix}, \begin{bmatrix} \mathbf{S}_b & \mathbf{0} & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{S}_d & \mathbf{0} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \boldsymbol{\Omega}_t^f & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{0} & \boldsymbol{\Omega}_t^u \end{bmatrix}\right) \tag{23}$$

where $\mathbf{S}_b = \mathrm{diag}(\mathbf{s}_b)$, $\mathbf{S}_d = \mathrm{diag}(\mathbf{s}_d)$.

References

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